CSE4203: Computer Graphics
Lecture - 3

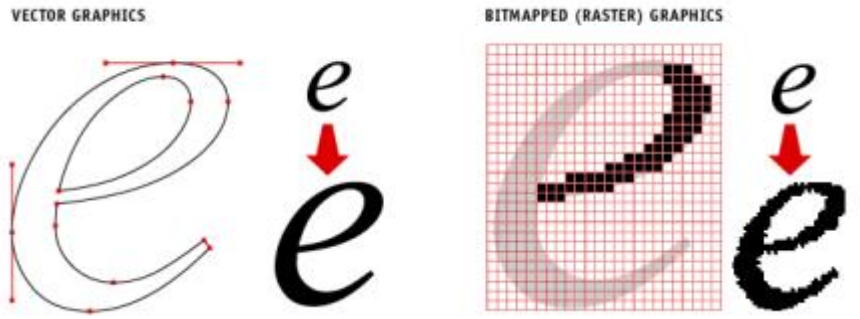
Vector Graphics

Outline

- Vector Image
- Curves
- Polynomial Curves
- Bézier Curves

Vector Image (1/3)

- storing descriptions of shapes
- areas of color bounded by lines or curves
- no reference to any pixel grid.



- Need to store *instructions for displaying the image rather* than the pixels needed to display it.

Vector Image (2/3)

- Resolution independent – scaling the image will not affect the quality
- Typically small in size compared to raster image
- Generates smooth curves and edges
- Limited photorealism
- Must be rasterized before they can be displayed
- File formats: SVG, AI, PDF

Vector Image (3/3)

- Differences between raster and vector images?

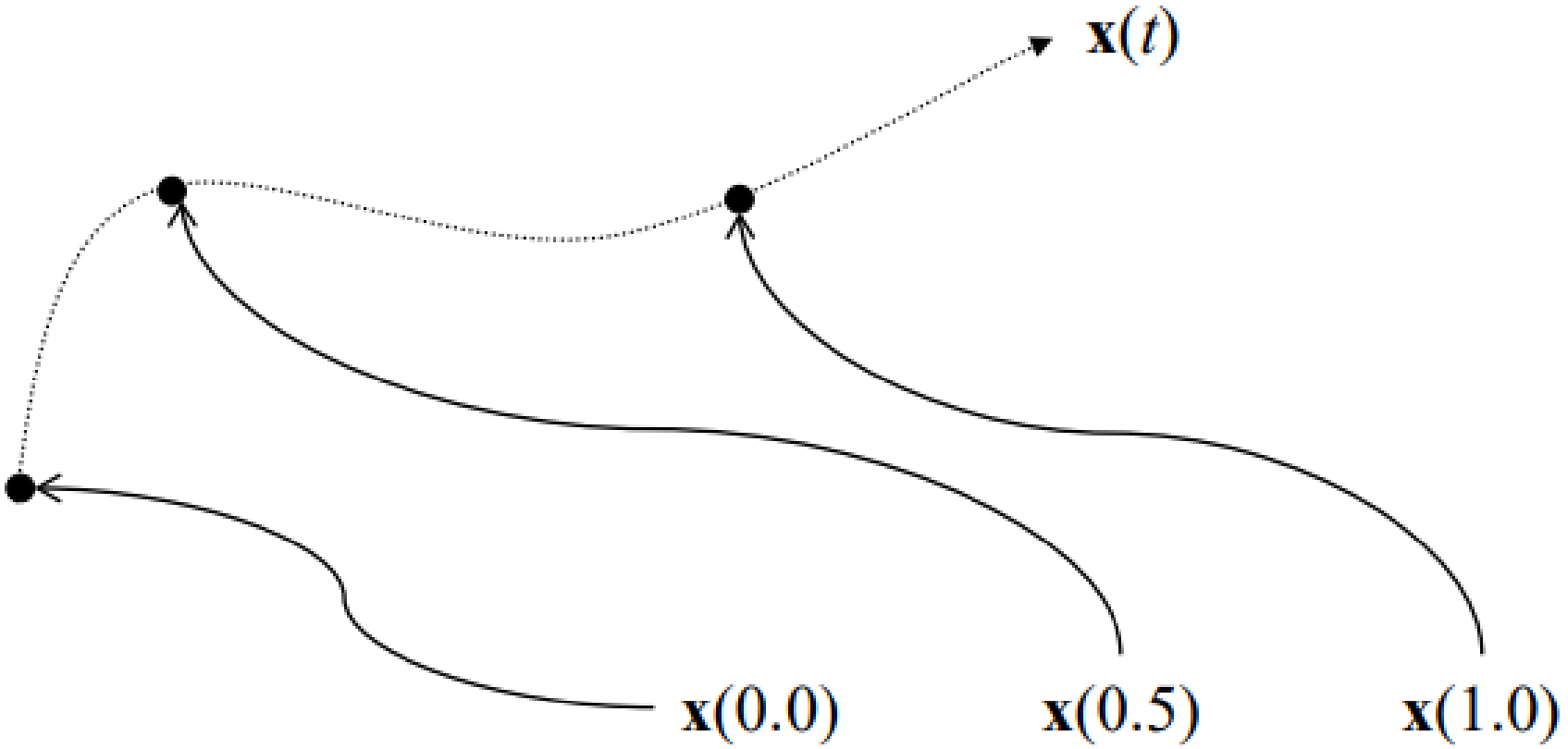
Curves (1/2)

- Curve is the continuous image of some interval in an n-dimensional space
- a continuous map from a one-dimensional space to an n-dimensional space.

Curves (2/2)

How many dimensions in the curve?

- It lies on 2D plane, but actually it's 1D



Credit: https://umd.instructure.com/courses/1286262/modules

Polynomial Curve (1/1)

- A polynomial is a sum of variables raised to powers and multiplied by coefficients

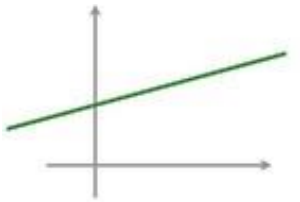
$$\sum(a_i x^i) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n$$

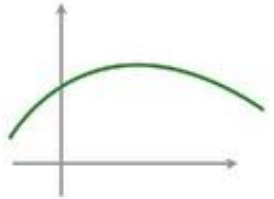
- The degree (or order) of a polynomial is the highest power of variables

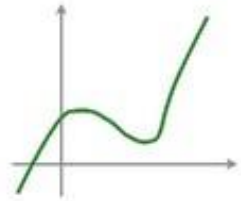
  1st degree (linear) polynomial: $y = ax^0 + bx^1$

  2nd degree (quadratic) polynomial: $y = ax^0 + bx^1 + cx^2$

  3rd degree (cubic) polynomial: $y = ax^0 + bx^1 + cx^2 + dx^3$







- Higher the degree, more change of directions

Bézier Curves (1/3)

- One of the most common representations for free-form curves in computer graphics
- First developed in 1959 by Paul de Casteljau
- Formalized and popularized by engineer Pierre Bézier
- Pierre Bézier used them for designing cars at Renault
- Common applications: 3D Modeling, CAD, typeface etc.



Pierre Étienne Bézier

Bézier Curves (2/3)

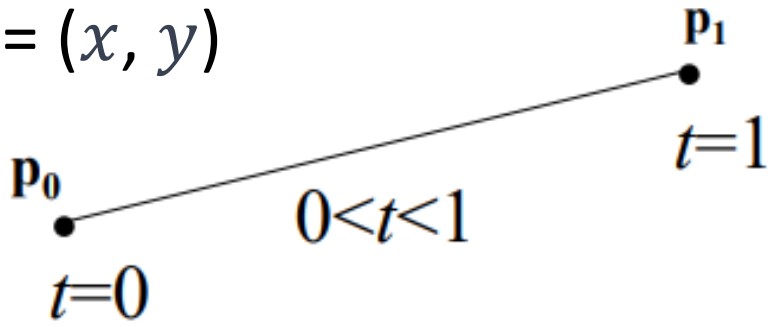
Parametric Equation

- Bézier Curves are expressed as parametric equations
- A parameter t is used to determine the value

$$x(t) = (1 - t)x_0 + tx_1$$
$$y(t) = (1 - t)y_0 + ty_1$$

where 0 ≤ t ≤ 1. Let $P_0 = (x_0, y_0)$, $P_1 = (x_1, y_1)$ and P = (x, y)

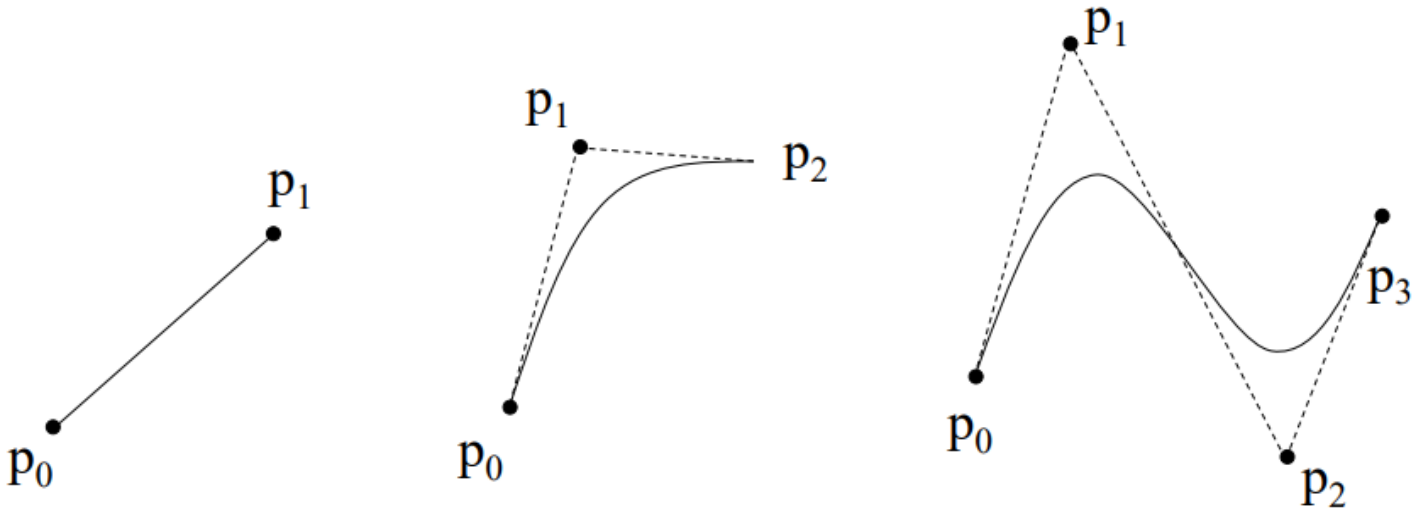
$$P(t) = (1 - t)P_0 + tP_1$$
or $P(t) = P_0 + t(P_1 - P_0)$



Bézier Curves (3/3)

Control Points

- A Bézier curve is defined by a set of control points
- An N degree Bézier curve is define by (N+1) control points
- For N control points, degree $d = N - 1$



Linear
N = 2

Quadratic
N = 3

Cubic
N = 4

Bézier Curves Derivation (1/13)

Derivation of a quadratic Bézier curve

- A quadratic (d=2) Bézier curve has 3 control points (P_0, P_1, P_2)
- Assume Q_0 and Q_1 lies on the line $P_0 \rightarrow P_1$ and $P_1 \rightarrow P_2$

$$Q_0 = P_0 + u(P_1 - P_0)$$

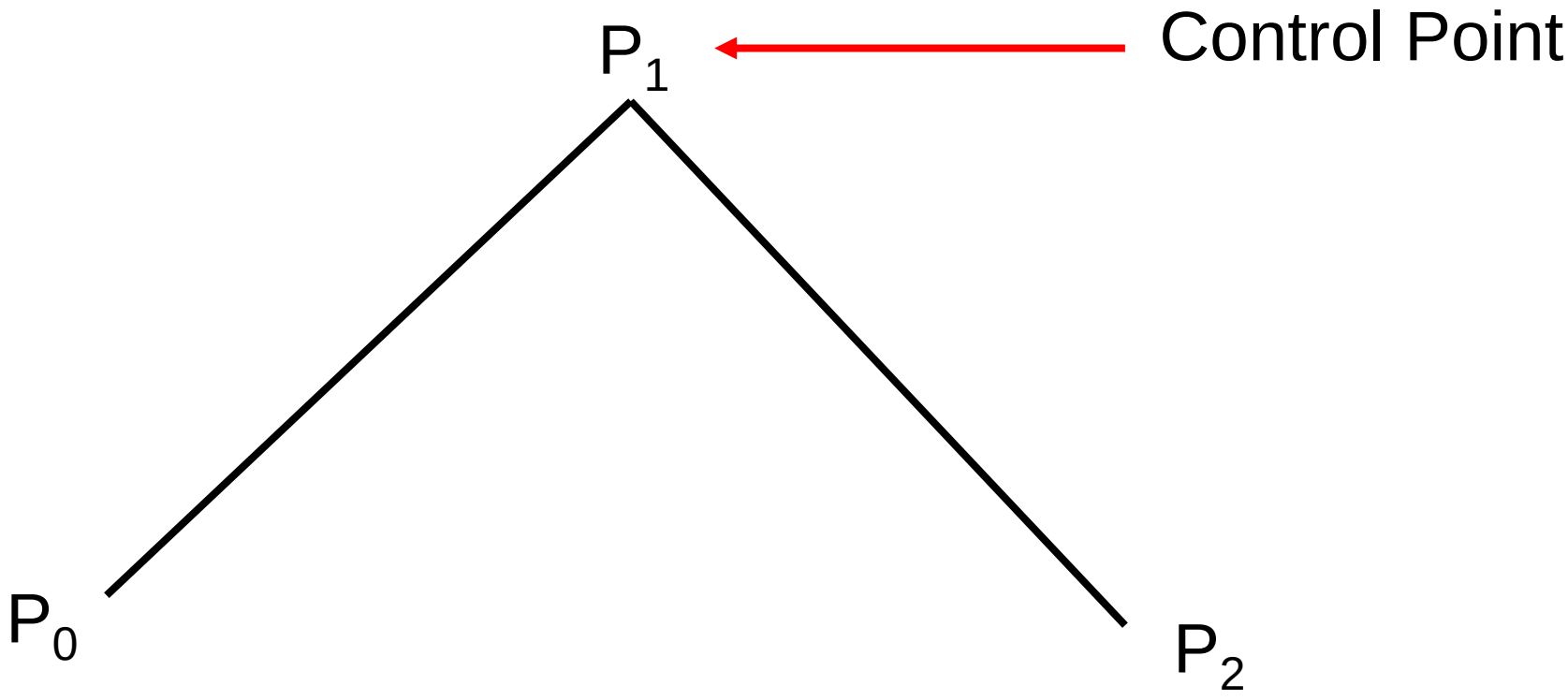
$$Q_1 = P_1 + u(P_2 - P_1)$$

- Point on the Bezier curve lies on the line $Q_0 \rightarrow Q_1$

$$Q(u) = Q_0 + u(Q_1 - Q_0)$$

Where u is a parameter ranges between 0 to 1

Bézier Curves Derivation (2/13)

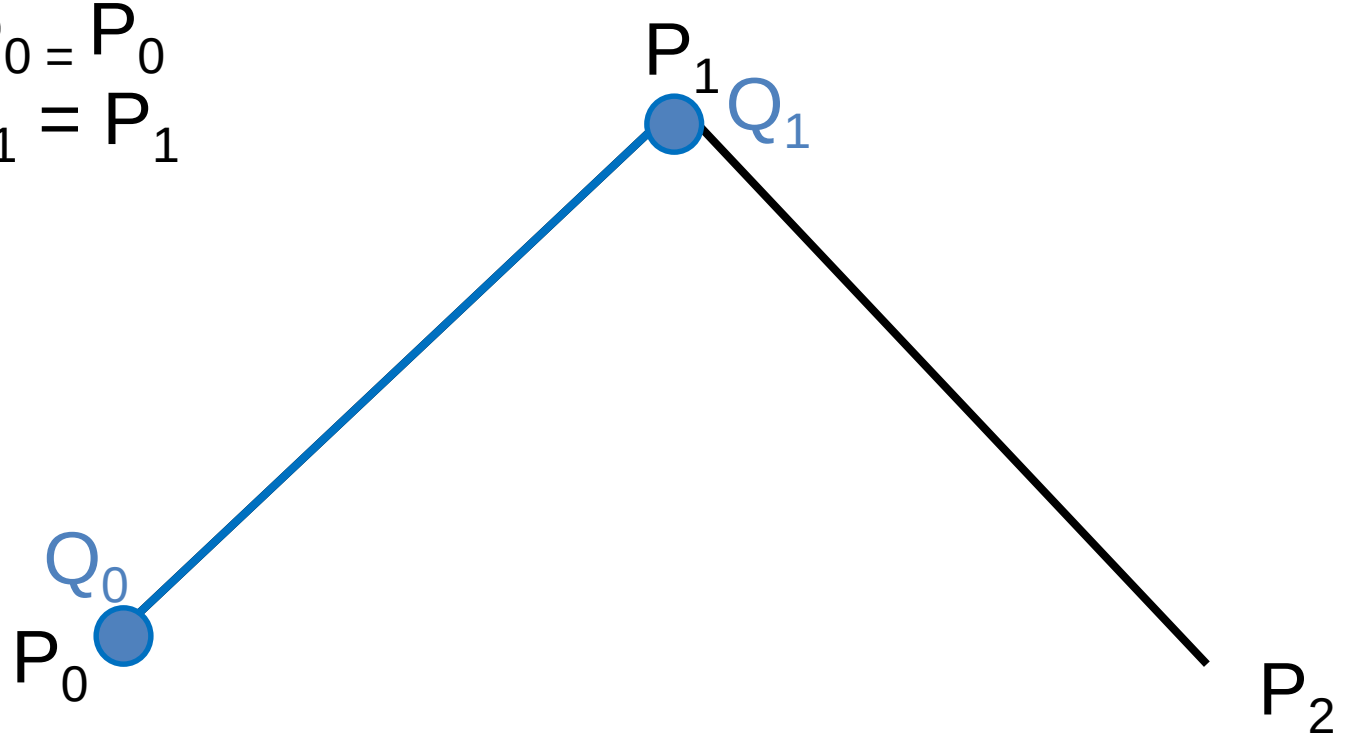


Bézier Curves Derivation (3/13)

at $u = 0$,

$Q_0 = P_0$

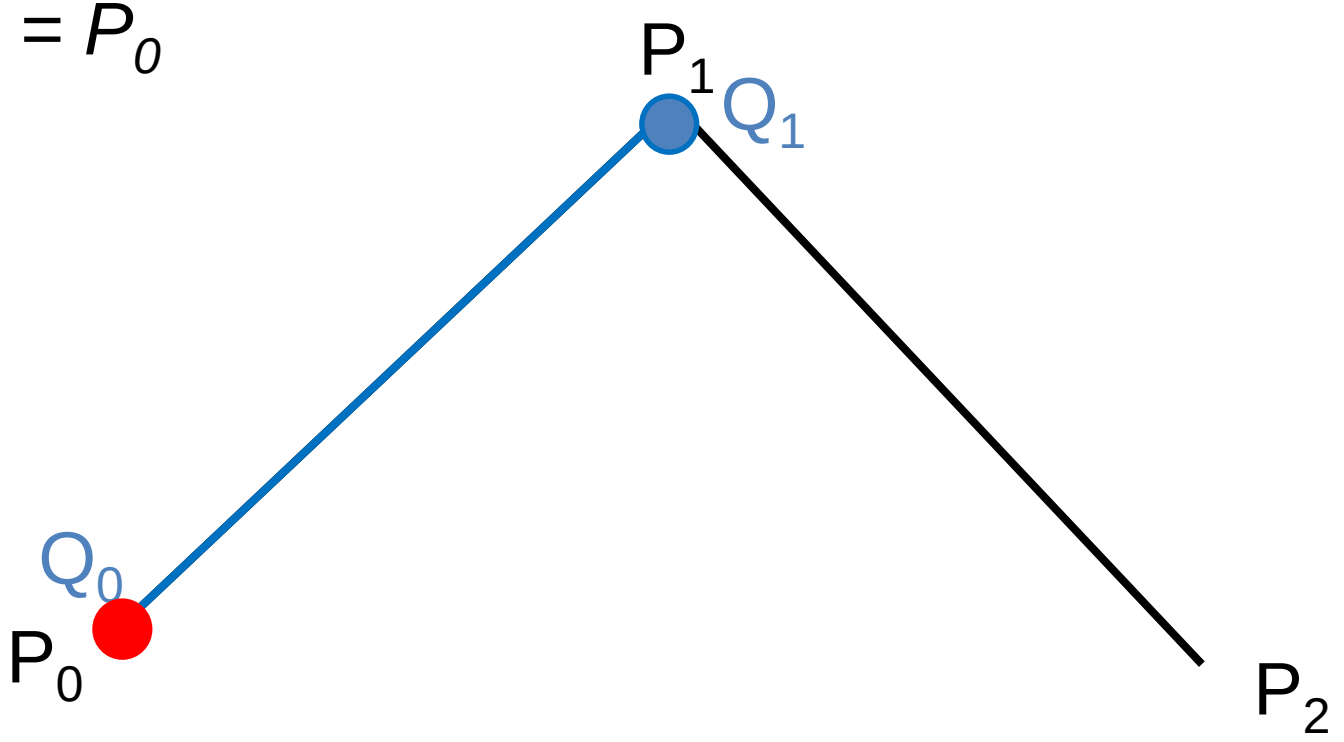
$Q_1 = P_1$



Bézier Curves Derivation (4/13)

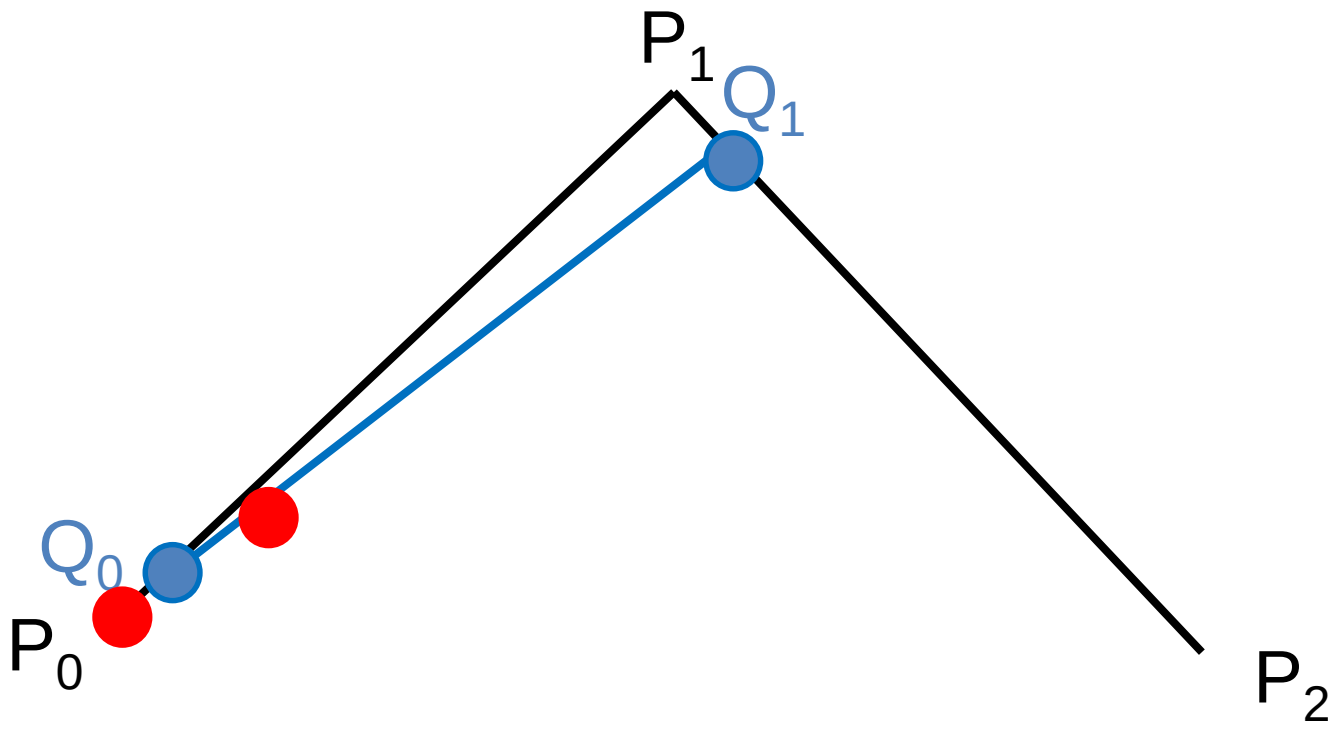
at $u = 0$,

$Q = P_0$



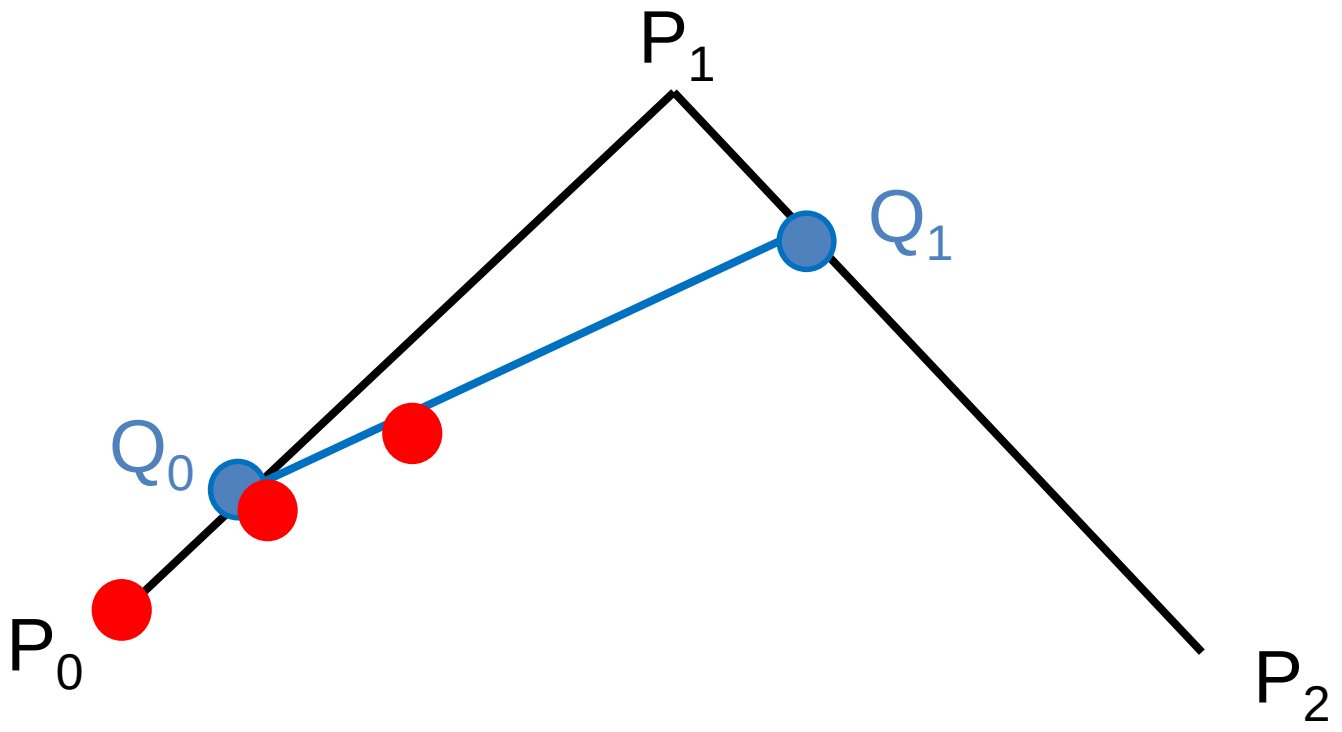
Bézier Curves Derivation (5/13)

at $u = 0.1$,



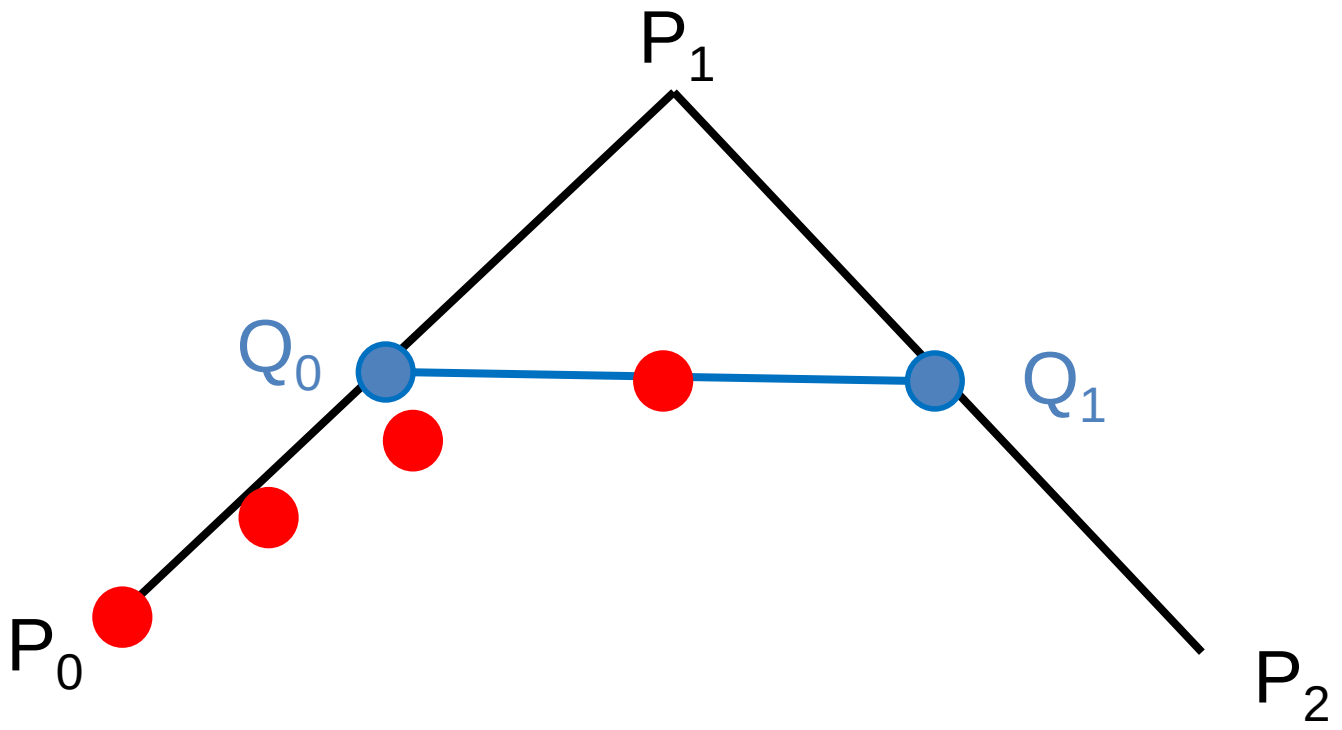
Bézier Curves Derivation (6/13)

at $u = 0.2$,



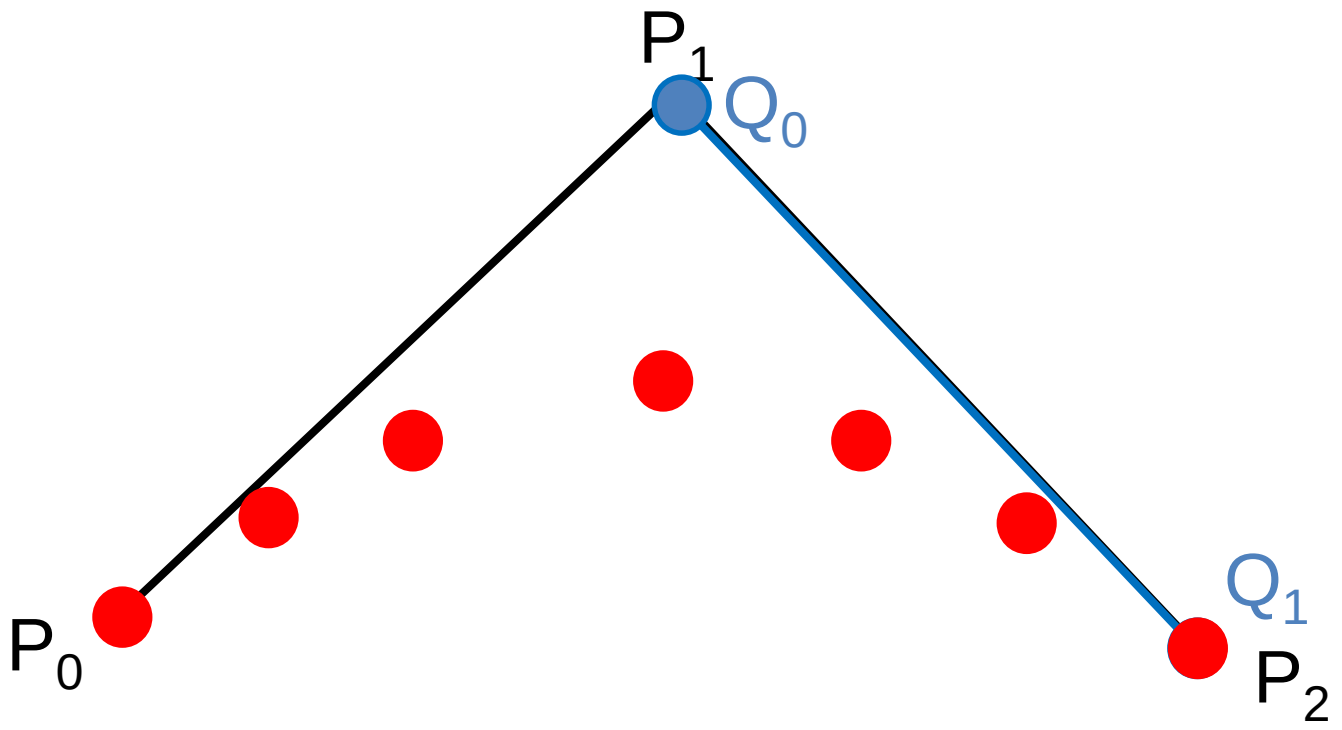
Bézier Curves Derivation (7/13)

at $u = 0.5$,

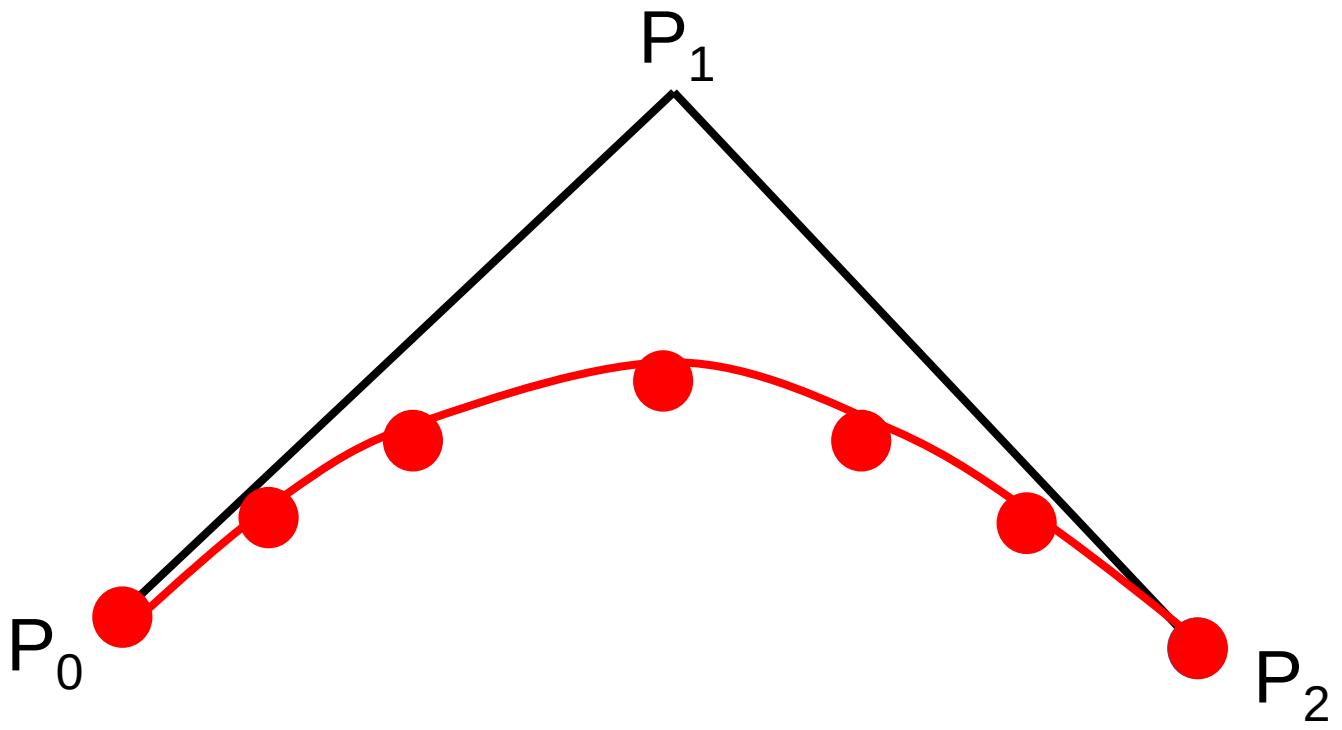


Bézier Curves Derivation (8/13)

at $u = 1$,



Bézier Curves Derivation (9/13)



Bézier Curves Derivation (10/13)

- Assume Q_0 and Q_1 lies on the line $P_0 \rightarrow P_1$ and $P_1 \rightarrow P_2$

$$Q_0 = P_0 + u(P_1 - P_0)$$

$$Q_1 = P_1 + u(P_2 - P_1)$$

- $Q(u)$ is the point on the Bezier curve on the line $Q_0 \rightarrow Q_1$

$$Q(u) = Q_0 + u(Q_1 - Q_0)$$

- Combining them gives

$$Q(u) = (1-u)^2 P_0 + 2u(1-u)P_1 + u^2 P_2$$

Bézier Curves Derivation (11/13)

de Casteljau's Algorithm

- This derivation is known is de Casteljau's Algorithm
- Let $P_{i,j}$ denote the control points
- where $P_{i,0}$ are the original control points P_0 to P_2

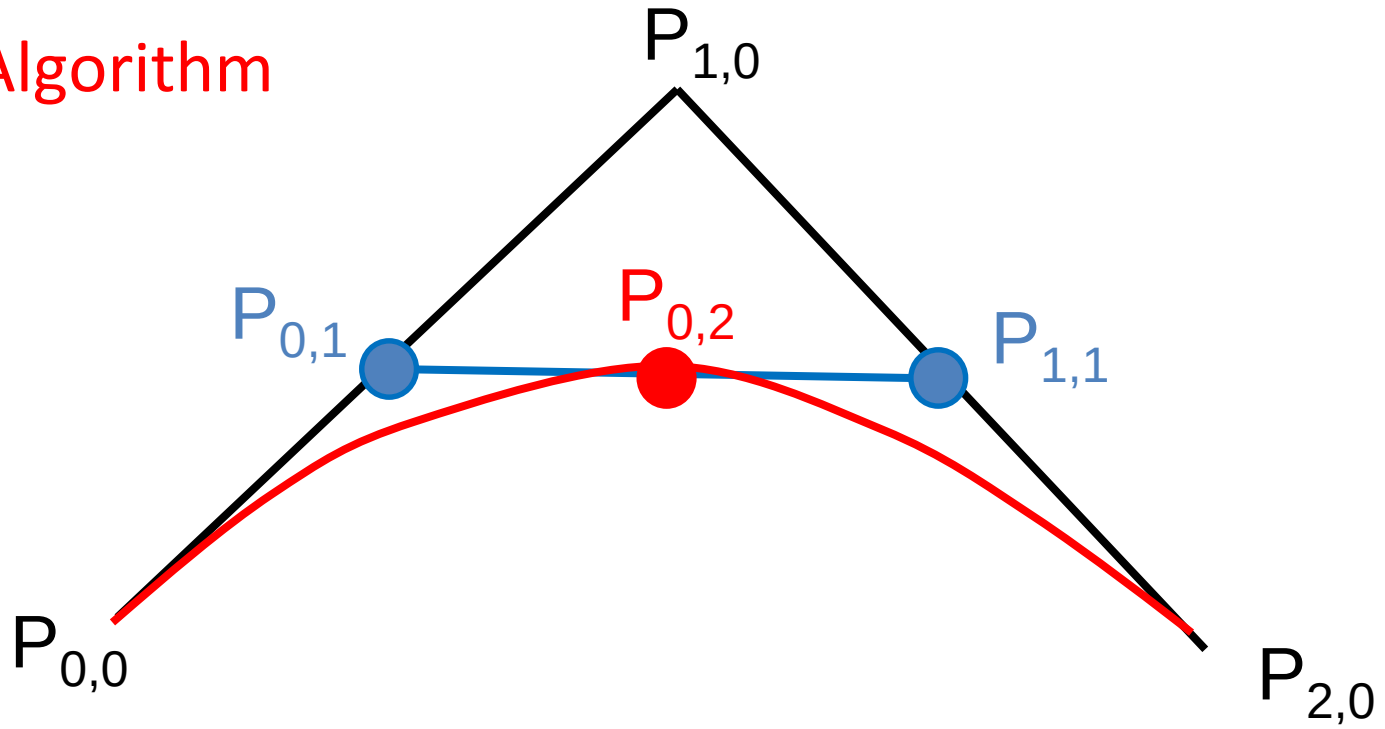
  $P_{i,1}$ are the points Q_1 to Q_2

  $P_{0,2}$ is the Q(u) then,

$$P_{i,j} = (1 - u)\, P_{i,j-1} + u\, P_{i+1,j-1}$$

Bézier Curves Derivation (12/13)

de Casteljau's Algorithm



$$P_{0,2} = (1-u)\,P_{0,1} + u\,P_{1,1}$$

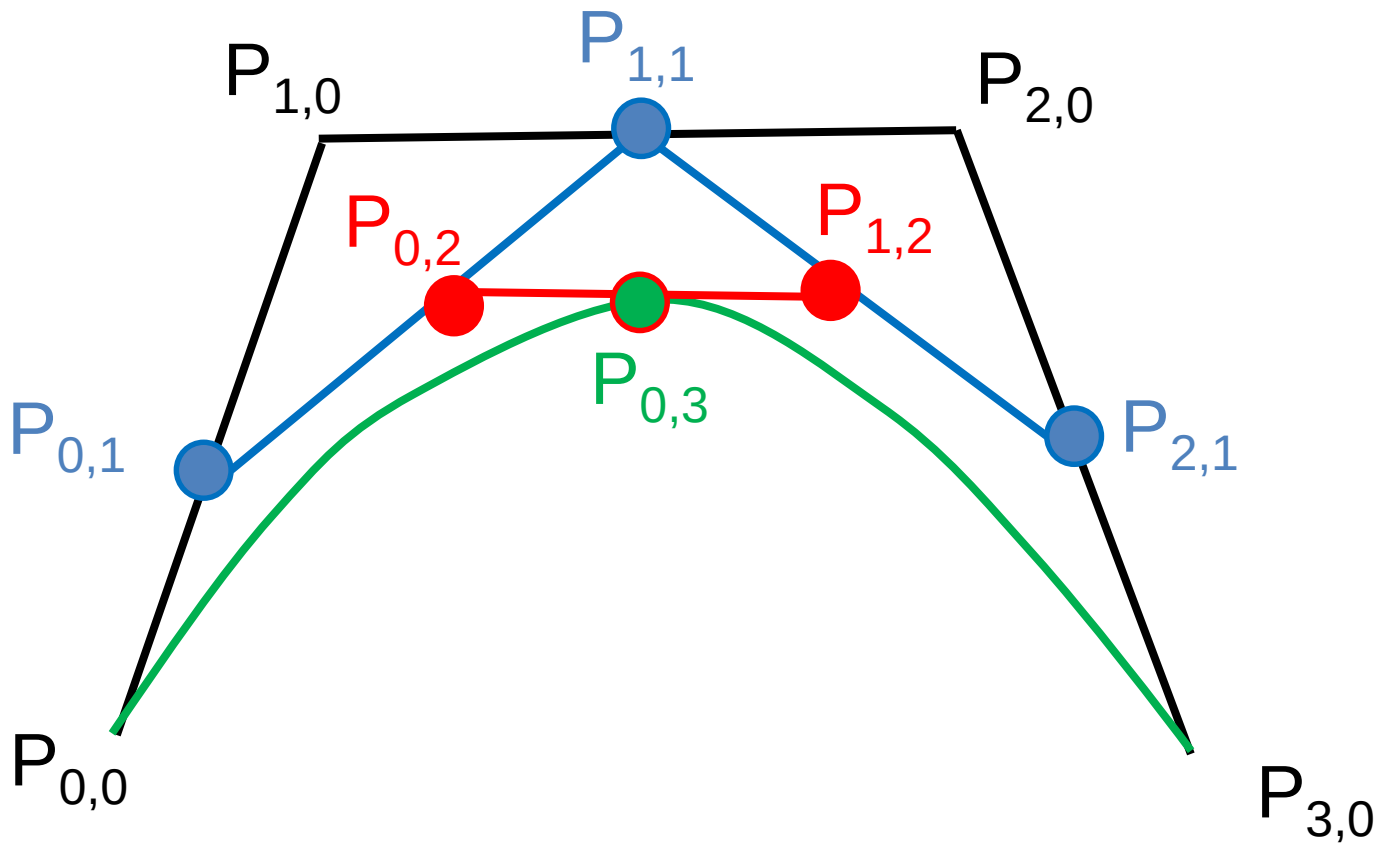
$$P_{0,2} = (1-u)\,[(1-u)\,P_{0,0} + u\,P_{1,0}] + u\,[(1-u)\,P_{1,0} + u\,P_{2,0}]$$

$$P_{0,2} = (1-u)^2\,P_{0,0} + 2u\,(1-u)\,P_{1,0} + u^2\,P_{2,0}$$

Bézier Curves Derivation (13/13)

For cubic Bézier Curves with 4 control points

$$P_{0,3} = (1-u)^3 P_{0,0} + 3u(1-u)^2 P_{1,0} + 3u^2(1-u) P_{2,0} + u^3 P_{3,0}$$



General form of Bézier Curves

General form of Bézier Curves:

$$Q(u) = \sum_{\mathrm{i}=0}^{\mathrm{d}} B_{i,d}(u)\mathbf{P}_i \qquad 0 \le u \le 1$$

$B_{i,d}(u)$ is called Bernstein polynomials

$$B_{\mathrm{i,d}}(u) = \binom{d}{i} u^i (1-u)^{d-i}$$

$$where \binom{d}{i} = \frac{d!}{i!(d-i)!} u^i (1-u)^{d-i}$$

Bézier curves are weighted sum of control points using n'th-order Bernstein polynomials

General form of Bézier Curves

General form of Bézier Curves:

$$Q(u) = \sum_{i=0}^{d} B_{i,d}(u)\mathbf{P}_i \qquad 0 \le u \le 1$$

$B_{i,d}(u)$ is called Bernstein polynomials

$$B_{i,d}(u) = \binom{d}{i} u^i (1-u)^{d-i}$$

$$where \binom{d}{i} = \frac{d!}{i!(d-i)!} u^i (1-u)^{d-i}$$

$$B_{0,2}(u) = (1-u)^2 \qquad B_{0,3}(u) = (1-u)^3$$
$$B_{1,2}(u) = 2u(1-u) \qquad B_{1,3}(u) = 3u(1-u)^2$$
$$B_{2,2}(u) = u^2 \qquad B_{2,3}(u) = 3u^2(1-u)$$
$$B_{3,3}(u) = u^3$$

$$Q_2(u) = (1-u)^2 P_0 + 2u(1-u) P_1 + u^2 P_2$$

General form of Bézier Curves

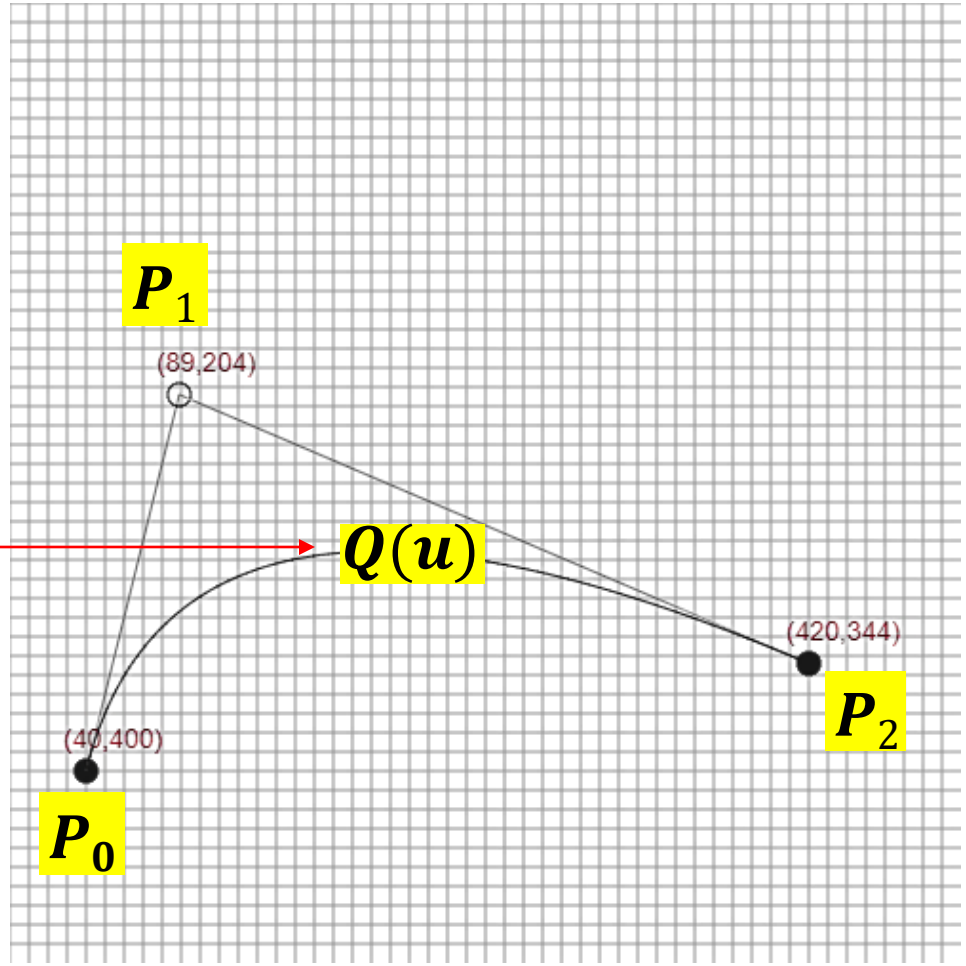
General form of Bézier Curves:

$$Q(u) = \sum_{i=0}^{d} B_{i,d}(u)\mathbf{P}_i$$

$$B_{i,d}(u) = \binom{d}{i} u^i (1-u)^{d-i}$$

$$where \binom{d}{i} = \frac{d!}{i!(d-i)!} u^i (1-u)^{d-i}$$

A point on the curve



General form of Bézier Curves

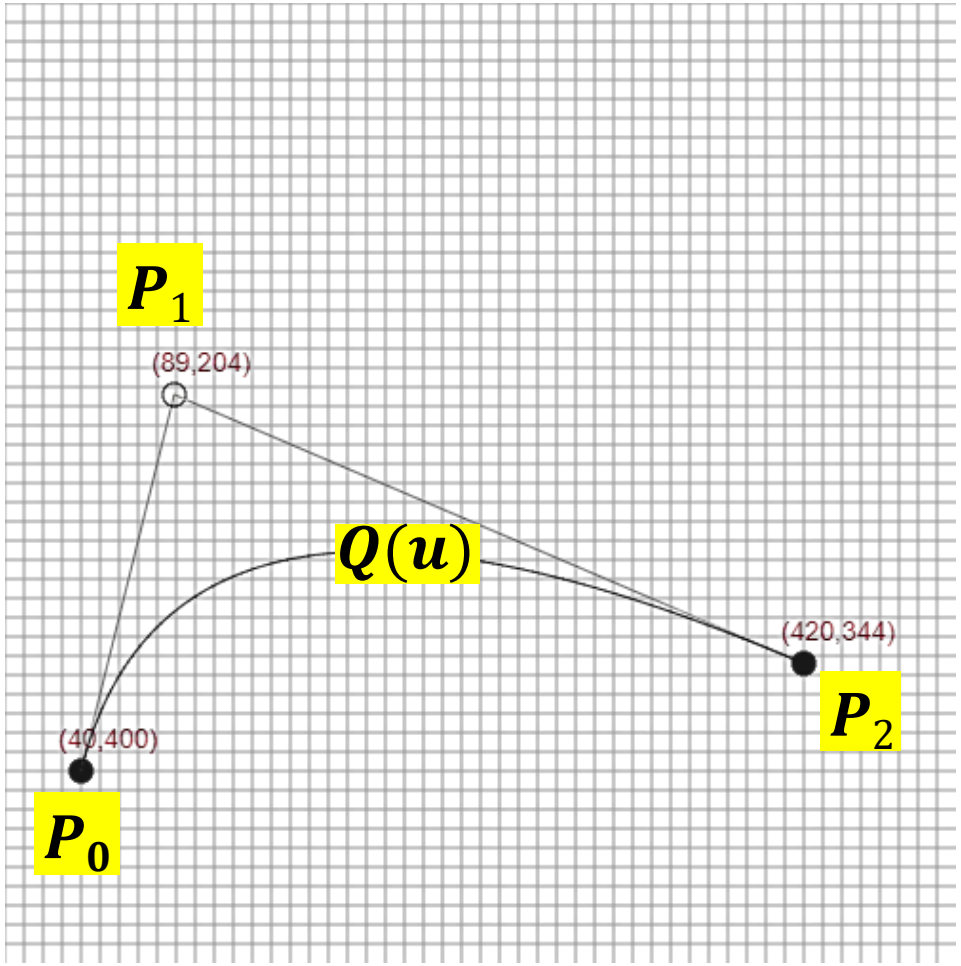
General form of Bézier Curves:

$$Q(u) = \sum_{\mathrm{i}=0}^{\mathrm{d}} B_{i,d}(u)\mathbf{P}_i$$

$$B_{\mathrm{i,d}}(u) = \binom{d}{i} u^i (1-u)^{d-i}$$

$$where \binom{d}{i} = \frac{d!}{i!(d-i)!} u^i (1-u)^{d-i}$$

Denoted with $Q_d(u)$



General form of Bézier Curves

General form of Bézier Curves:

$$Q(u) = \sum_{i=0}^{d} B_{i,d}(u)\mathbf{P}_i$$

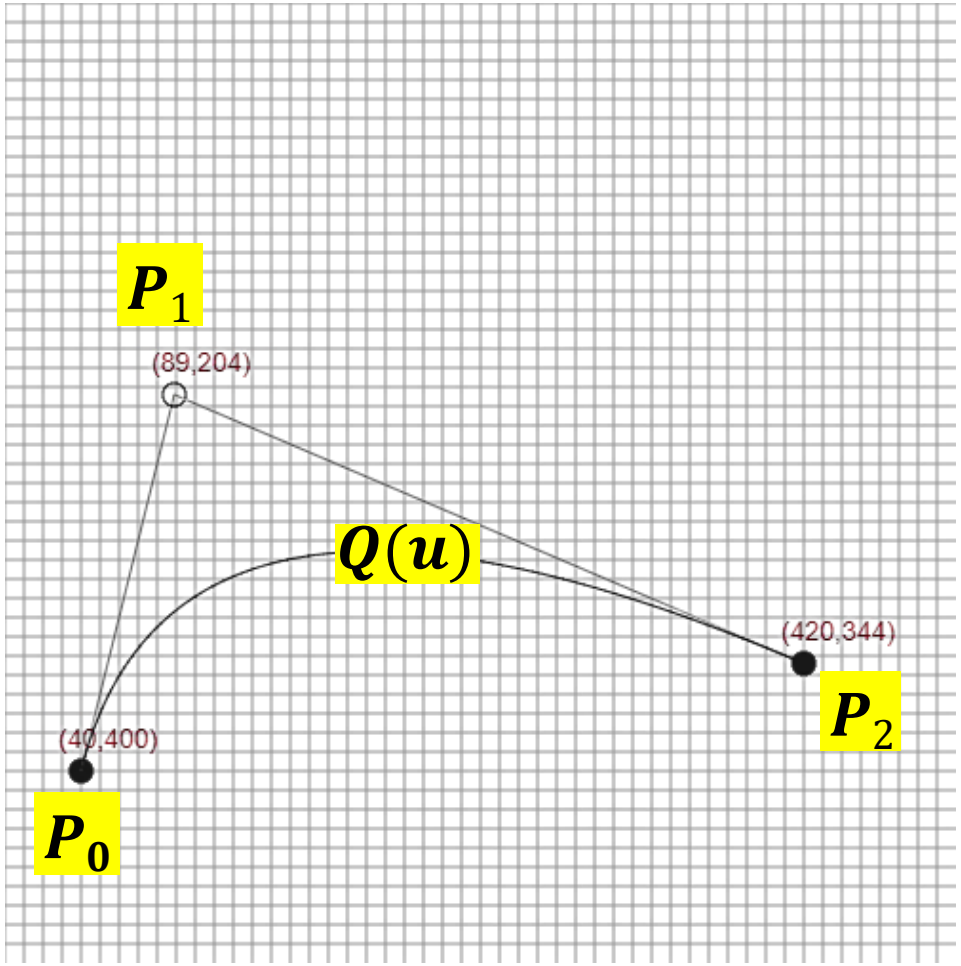
$$B_{i,d}(u) = \binom{d}{i} u^i (1-u)^{d-i}$$

$$where \binom{d}{i} = \frac{d!}{i!(d-i)!} u^i (1-u)^{d-i}$$

Where is $Q_d(0.5)$ situated?

Where is $Q_d(0)$ situated?

Where is $Q_d(1)$ situated?



Practice Problem 1

Given control points $P_0 = (0, 0), P_1 = (4, 2), P_2 = (8, 0)$, find the Bézier curve values $Q_2(0)$, $Q_2(\frac{1}{2})$ and $Q_2(1)$.

Why subscript 2 for $Q_2(u)$?

Practice Problem 1

Given control points $P_0 = (0,0), P_1 = (4,2), P_2 = (8,0)$, find the Bézier curve values $Q_2(0)$, $Q_2(\frac{1}{2})$ and $Q_2(1)$.

$$Q_2(u) = \sum_{i=0}^{n} B_{i,2}(u) P_i \qquad 0 \le u \le 1$$

$$B_{i,d}(u) = \begin{pmatrix} d \\ i \end{pmatrix} u^i (1-u)^{d-i} \qquad \begin{pmatrix} d \\ i \end{pmatrix} = \frac{d!}{i!(d-i)!}$$

$$Q_2(u) = B_{0,2}(u)P_0 + B_{1,2}(u)P_1 + B_{2,2}(u)P_2$$

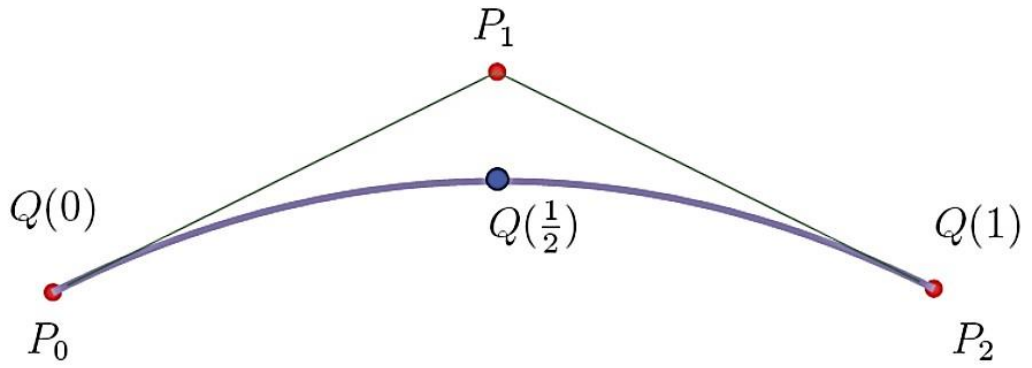
$$Q_2(u) = (1-u)^2 P_0 + 2(1-u)uP_1 + u^2 P_2$$

Practice Problem 1

Given control points $P_0 = (0,0), P_1 = (4,2), P_2 = (8,0)$, find the Bézier curve values $Q_2(0)$, $Q_2(\frac{1}{2})$ and $Q_2(1)$.

$$Q_2(u) = (1-u)^2 P_0 + 2(1-u)uP_1 + u^2 P_2$$

- $Q_2(0) = (1-0)^2 P_0 + 2(1-0)0P_1 + 0^2 P_2 = P_0 = (0,0)$
- $Q_2(\frac{1}{2}) =$ *….. Do calculations ….* $= (4,1)$
- $Q(1) =$ *… Do calculations …* $= (8,0)$

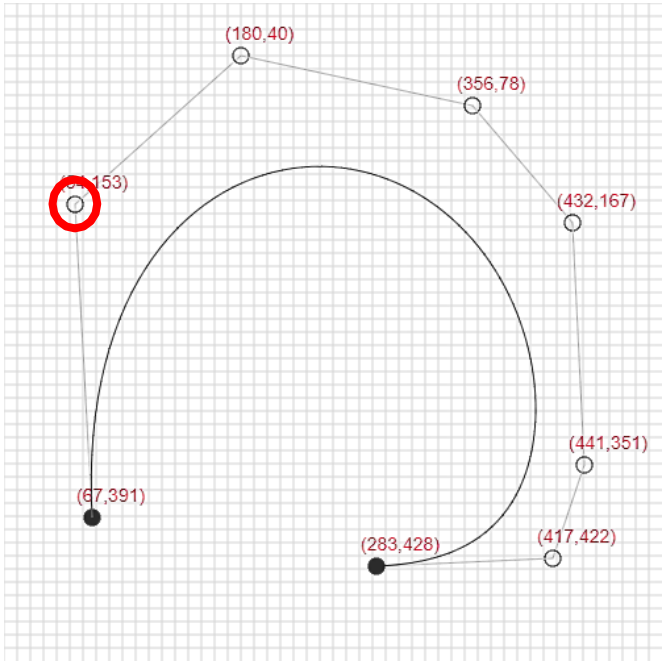


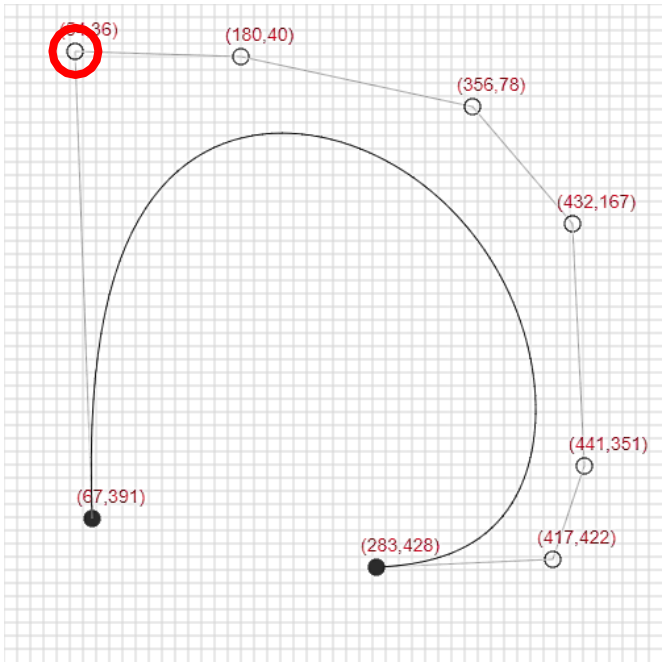
Properties of Bezier Curves

- They generally follow the shape of the control polygon, which consists of the segments joining the control points
- They always pass through the first and last control points
- They are contained in the convex hull of their defining control points
- The degree of the polynomial defining the curve segment (d) is one less than that the number of defining polygon point (n) i.e. n = d+1

Disadvantages

- A change to any of the control point alters the entire curve.
- Having a large number of control points requires high polynomials to be evaluated. This is expensive to compute.





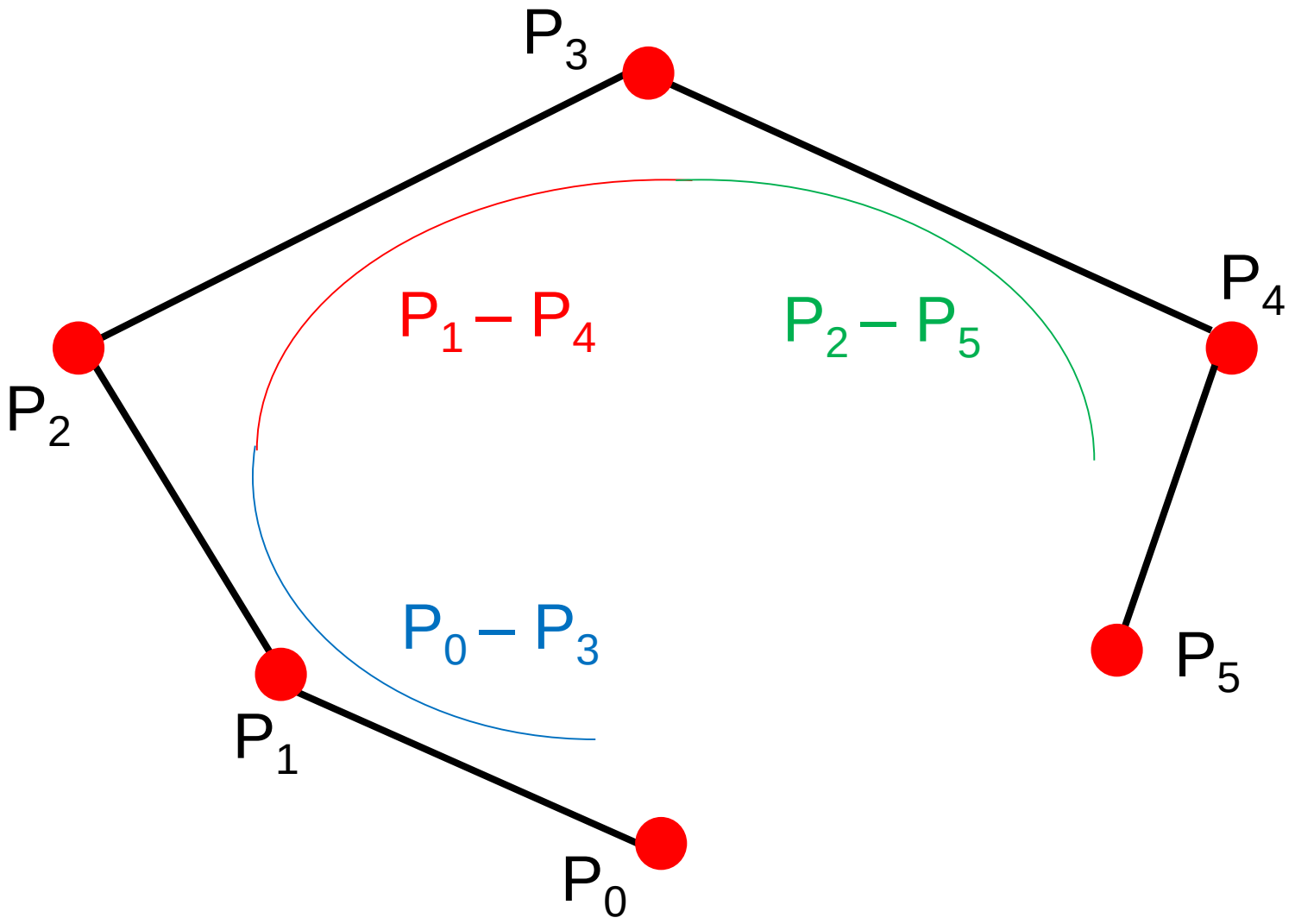
Credit: CPSC 589/689 Course Notes, University of Calgary, Faramarz Samavati

B-Spline Curve (1/4)

- B-splines (or Basis Splines) use several Bezier Curves joined end on end
- a k degree B-spline curve defined by n+1 control points will consists of n – k + 1 Bezier Curves
- For example, a cubic B-Spline Curve defined by 6 control points P_0 P_1 P_2 P_3 P_4 P_5 consists of n – k + 1 = 5 – 3 +1 = 3 Bezier Curves

Credit: Mathematics of Computer Graphics and Virtual Environments – Dr. Jon Siach

B-Spline Curve (2/4)



Cubic B-Spline Curve with 6 control Points

B-Spline Curve (3/4)

- Degree is independent of the number of control points
- The final point of the first Bezier curve has the same co-ordinate as the first point on the second Bezier curve (C^0 continuity)
- The first derivative at the end of the first Bezier curve is the same as the first derivative at the start of the second Bezier curve (C^1 continuity)
- The second derivative at the end of the first Bezier curve is the same as the second derivative at the start of the second Bezier curve (known as C2 continuity).

Credit: Mathematics of Computer Graphics and Virtual Environments – Dr. Jon Siach

B-Spline Curve (4/4)

- A B-spline curve S(t), is defined by,

$$S(t) = \sum_{i=0}^{n} N_{i,k}(t)\mathbf{P}_i$$

- where (P_0, P_1,...,P_n) are the control points
- k is the order of the polynomial segments of the B-spline curve
- $N_{i,k}(t)$ are the “normalized B-spline blending functions”.

Credit: Mathematics of Computer Graphics and Virtual Environments – Dr. Jon Siach

Cox-de Bour formula (1/1)

- The blending function $N_{i,k}(t)$ is defined by Cox-de Bour recursion formula,

$$N_{i,0}(t) = \begin{cases} 1 & \text{if } t_i \leq t < t_{i+1} \\ 0 & \text{otherwise} \end{cases},$$

$$N_{i,j}(t) = \frac{t - t_i}{t_{i+j} - t_i} N_{i,j-1}(t) + \frac{t_{i+j+1} - t}{t_{i+j+1} - t_{i+1}} N_{i+1,j-1}(t).$$

- The values of t_i is taken from non-decreasing sequence of real numbers called knot vector, $T = \{t_0, t_1, t_2, \ldots\ldots, t_m\}$
- The number of knots in a knot vector, $m = k + n + 1$

Credit: Mathematics of Computer Graphics and Virtual Environments – Dr. Jon Siach

Uniform Quadratic B-spline (1/4)

- If the knots are equidistant then we have a uniform B-spline
- If a uniform quadratic B-spline is defined by the control points (P_0, P_1, P_2), hence k = 2, n = 2
- then m = k + n + 1= 2 + 2 + 1 = 5
- knot vector, T = {t_0, t_1, t_2, t_3, t_4, t_5} = {0, 1, 2, 3, 4, 5}
- The B-Spline curve is defined by,

$$P(t) = \sum_{i=0}^{2} N_{i,2}(t)\mathbf{P}_i$$

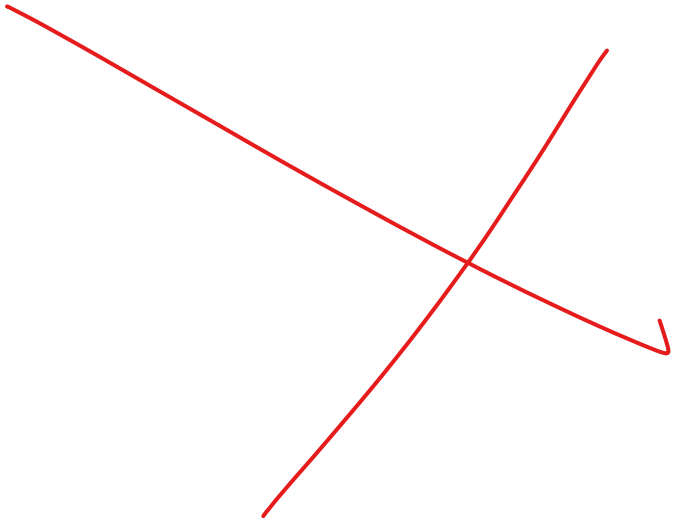
Uniform Quadratic B-spline (2/4)

- Equation of a quadratic B-spline curve is

$$P(t) = \sum_{i=0}^{n} N_{i,k}(t)\mathbf{P}_i$$

- when $t_2 \leq t \leq t_3$

$$N_{0,2}(t)P_0 = \frac{1}{2}[(t-t_2)^2 - 2(t-t_2) + 1]P_0,$$
$$N_{1,2}(t)P_1 = \frac{1}{2}[-2(t-t_2)^2 + 2(t-t_2) + 1]P_1,$$
$$N_{2,2}(t)P_2 = \frac{1}{2}(t-t_2)^2 P_2,$$



Credit: Mathematics of Computer Graphics and Virtual Environments – Dr. Jon Siach

Uniform Quadratic B-spline (3/4)

- which can be written in matrix form as:

$$S(t) = \begin{pmatrix} P_0 & P_1 & P_2 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & -2 & 1 \\ -2 & 2 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} t^2 \\ t \\ 1 \end{pmatrix}$$

- In general terms a uniform quadratic B-spline curve is written as:

$$S_i(t) = \begin{pmatrix} P_i & P_{i+1} & P_{i+2} \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & -2 & 1 \\ -2 & 2 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} t^2 \\ t \\ 1 \end{pmatrix},$$

where i = 0, 1,...,n − k + 1

Credit: Mathematics of Computer Graphics and Virtual Environments – Dr. Jon Siach

Uniform Quadratic B-spline (4/4)

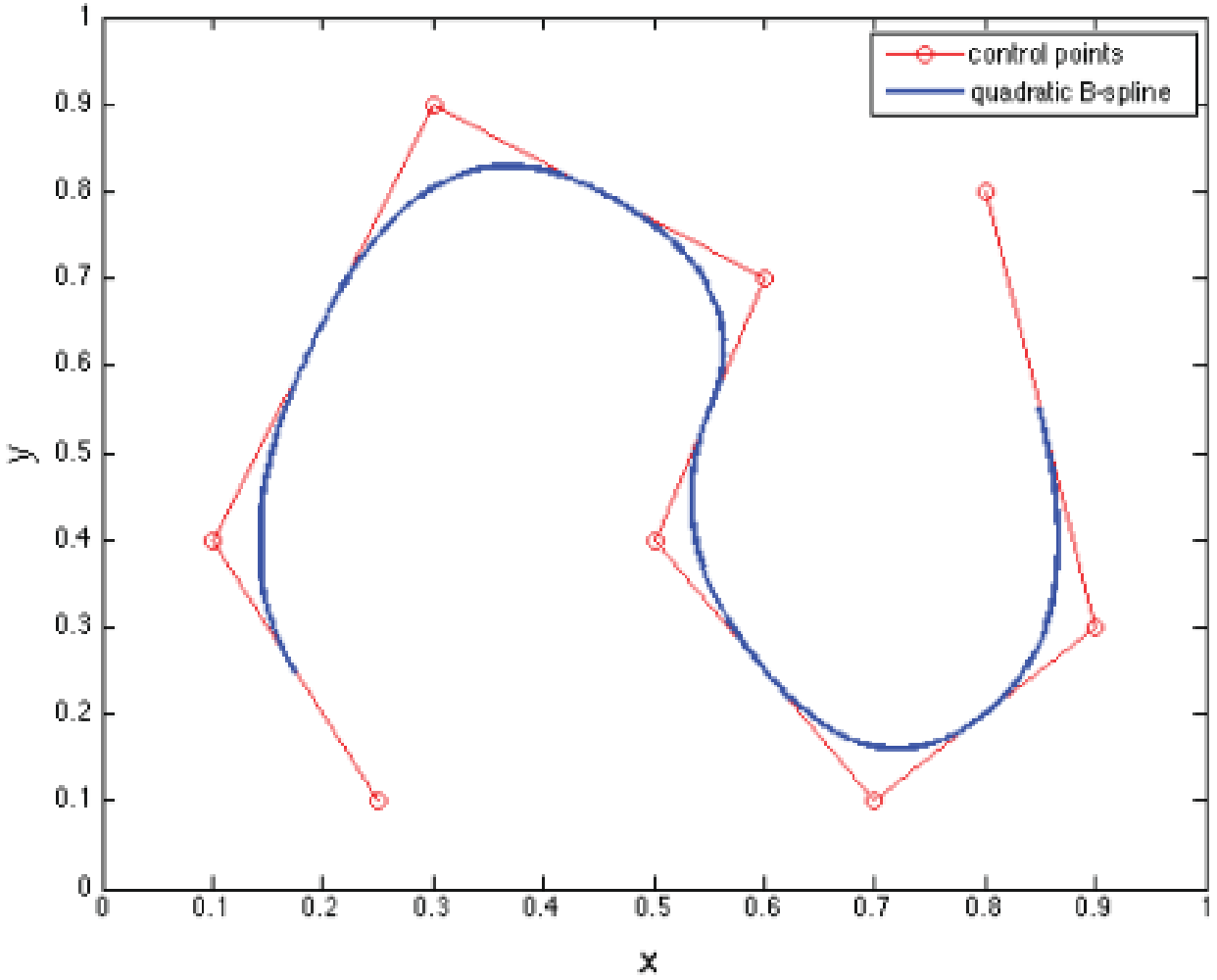


Fig: A uniform quadratic B-spline curve

Credit: Mathematics of Computer Graphics and Virtual Environments – Dr. Jon Siach

Open Uniform B-spline curves (1/3)

- Uniform B-splines aren't defined for all the range $[t_0, t_m]$.
- As a result, B-spline curve is undefined at the start and end control points.
- Open uniform B-spline curves overcome this by setting the first k knots the same value and the last k+1 knots have the same value
- For example, consider an open uniform quadratic B-spline curve defined by the control points P_0, …., P_4. Here m = k +n + 1 = 2 + 4 + 1 = 7
- So the knot vector is T = $\{t_0, t_1, t_2, t_3, t_4, t_5, t_6, t_7\}$

  = {0, 0, 0, 1, 2, 3, 3, 3}

Credit: Mathematics of Computer Graphics and Virtual Environments – Dr. Jon Siach

Open Uniform B-spline curves (2/3)

- Matrix form of an open uniform B-spline curve is

$$S_0(t) = \begin{pmatrix} P_0 & P_1 & P_2 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 2 & -4 & 2 \\ -3 & 4 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} t^2 \\ t \\ 1 \end{pmatrix},$$

$$S_i(t) = \begin{pmatrix} P_i & P_{i+1} & P_{i+2} \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & -2 & 1 \\ -2 & 2 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} t^2 \\ t \\ 1 \end{pmatrix}, \qquad 1 \leq i \leq n-3,$$

$$S_{n-2}(t) = \begin{pmatrix} P_{n-2} & P_{n-1} & P_n \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 & -2 & 1 \\ -3 & 2 & 1 \\ 2 & 0 & 0 \end{pmatrix} \begin{pmatrix} t^2 \\ t \\ 1 \end{pmatrix},$$

Credit: Mathematics of Computer Graphics and Virtual Environments – Dr. Jon Siach

Open Uniform B-spline curves (3/3)

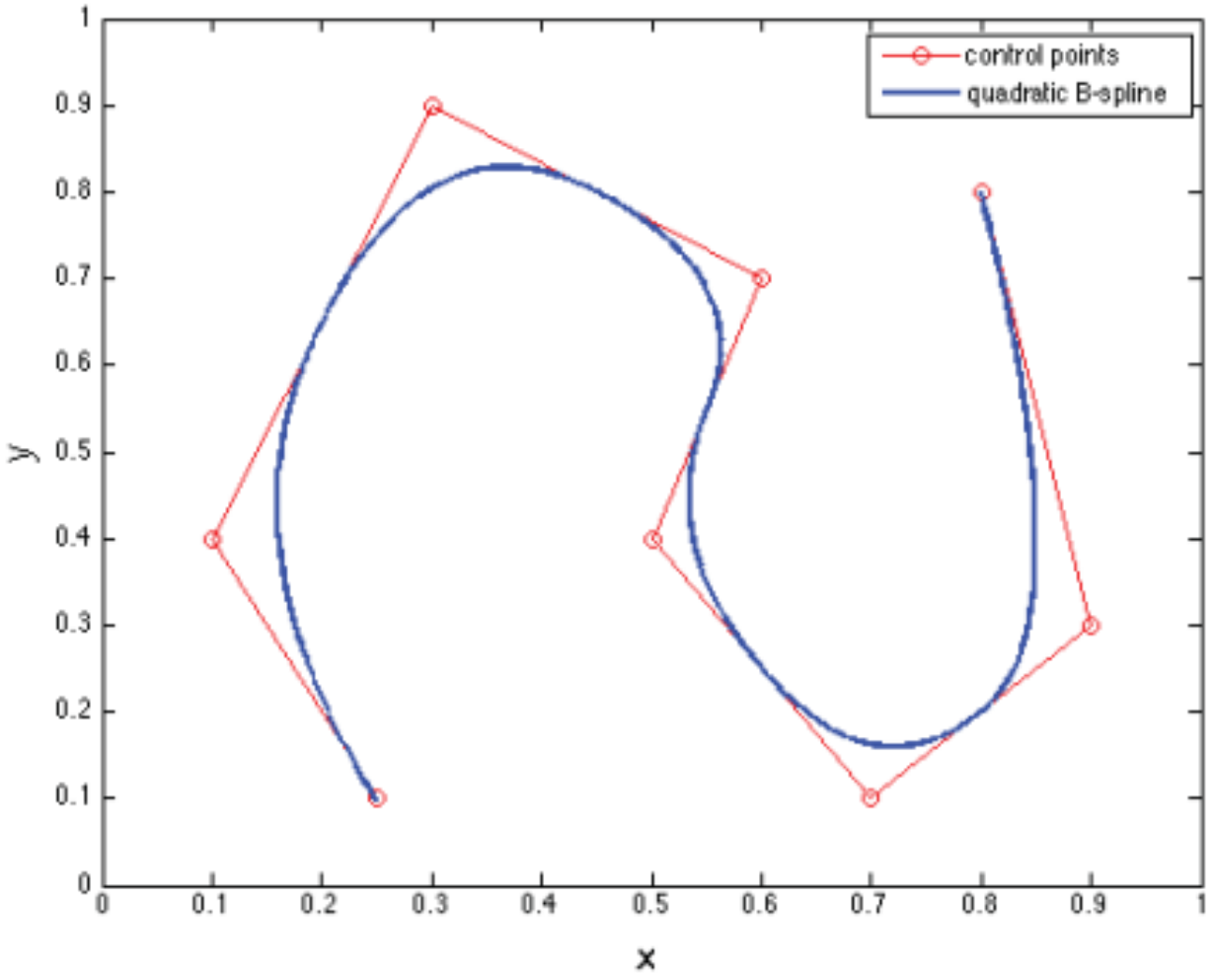


Fig: A Open Uniform B-spline curve

Credit: Mathematics of Computer Graphics and Virtual Environments – Dr. Jon Siach

Practice Problem 2

An uniform quadratic B-Spline curve is defined by 5 control points $P_0(1, 2)$, $P_1(3, 8)$, $P_2(5, 2)$, $P_3(7, 1)$ and $P_4(8, 0)$. Find the points on the curve segments for t = 0

Practice Problem 2

An uniform quadratic B-Spline curve is defined by 5 control points $P_0(1, 2)$, $P_1(3, 8)$, $P_2(5, 2)$, $P_3(7, 1)$ and $P_4(8, 0)$. Find the points on the curve segments for t = 0

solution:

k = 2, n = 4

Number of curve segments = n – k + 1 = 4 – 2 + 1 = 3

$$S_i(0) = \begin{bmatrix} P_i & P_{i+1} & P_{i+2} \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 2 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} t^2 \\ t \\ 1 \end{bmatrix}$$

Practice Problem 2

$$S_0(0) = \begin{bmatrix} P_0 & P_1 & P_2 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 2 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} t^2 \\ t \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 8 & 2 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 2 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$S_1(0) = \begin{bmatrix} P_1 & P_2 & P_3 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 2 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} t^2 \\ t \\ 1 \end{bmatrix} = ?$$

$$S_2(0) = \begin{bmatrix} P_2 & P_3 & P_4 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 2 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} t^2 \\ t \\ 1 \end{bmatrix} = ?$$

Credit: Mathematics of Computer Graphics and Virtual Environments – Dr. Jon Siach

Practice Problem 3

An open uniform quadratic B-Spline curve is defined by 5 control points $P_0(1, 2)$, $P_1(3, 8)$, $P_2(5, 2)$, $P_3(7, 1)$ and $P_4(8, 0)$. Find the points on the curve segments for t = 0

solution:

k = 2, n = 4

Number of curve segments = n – k + 1 = 4 – 2 + 1 = 3

Practice Problem 3

$$S_0(0) = \begin{bmatrix} P_0 & P_1 & P_2 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 2 & -4 & 2 \\ -3 & 4 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} t^2 \\ t \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 8 & 2 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 2 & -4 & 2 \\ -3 & 4 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$S_1(0) = \begin{bmatrix} P_1 & P_2 & P_3 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 2 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} t^2 \\ t \\ 1 \end{bmatrix} = ?$$

$$S_2(0) = \begin{bmatrix} P_2 & P_3 & P_4 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & -2 & 1 \\ -3 & 2 & 1 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} t^2 \\ t \\ 1 \end{bmatrix} = ?$$

Credit: Mathematics of Computer Graphics and Virtual Environments – Dr. Jon Siach

Further Reading

- Fundamentals of Computer Graphics, 4th Edition - Chapter 15
- https://www.youtube.com/watch?v=2HvH9cmHbG4
- https://www.youtube.com/watch?v=qhQrRCJ-mVg

End

Credit: Mathematics of Computer Graphics and Virtual Environments – Dr. Jon Siach